Wordle Lab: Take-home Exercise

The task

Build a small web app where a user can play Wordle, and where a solver can suggest the best next guess and explain why.

Use any language, framework, or tools you like, including AI assistants. We care more about clarity and correctness than polish.

What to build

1. Game engine

- Score a guess against an answer (green / yellow / gray)
- Track game state: guesses so far, win or lose, and whether a word is allowed
- Load the word list from a file

2. Solver

- Given the guesses and results so far, work out which words are still possible
- Rank possible next guesses by expected information (the entropy of the result pattern across the remaining words)
- Return the top suggestions with their scores

3. HTTP API

- Start a game
- Submit a guess and get back the result and game status
- Request a hint: the number of remaining words and the top suggestions with their entropy
- Return clear error messages for invalid words or a finished game

4. Web page

- A playable board with a keyboard and colored tiles
- A hint panel showing suggestions, their entropy, and how many words remain
- A "words remaining" counter that updates after each guess
- Graceful handling of errors, without breaking the page

Optional stretch

Add a "simulate" feature that runs your solver against every answer in the list and shows how many guesses it needed. Only attempt this if you have time left.

Ground rules

- No login, database, or deployment is needed. A project that runs locally with clear instructions is the finish line.

- Keep the game logic separate from the web code, so the same logic works whichever way it is used.
- Be ready to tell us which parts an AI tool wrote and how you checked them.

What to send us

A link to a repository or a zip file containing the code, tests, and README. Please include instructions we can follow to run it.